

Birthday attack- derivation

Module IV



The Birthday paradox

- Hash functions have collisions as a result of the basic hash function requirements. The first requirement specifies that $h(x)$ must accept inputs of arbitrary size.
- The second requirement specifies that $h(x)$ must generate an output of fixed size.
- Therefore, the size of the Input message space is significantly larger than the size of the output hashed value space.
- As a result, there must exist multiple messages that map to the same hashed value.
- The nature of collisions is illustrated with a phenomenon known as the Birthday Paradox.
- Consider a question, “how many people must be present so that there is a 50% probability that at least two of the people have the same birth date.” (Ignore leap year).
- There are 365 possible birth dates.

The Birthday paradox cntd..

- The probability that a person has a given birth date is $\frac{1}{365}$.
- Thus, the probability that no two people do not have the same birth date, i.e. that there is no collision is $1 - \frac{1}{365}$.
- In the case, where there are three people, the probability that no two people have the same birth date is:

$$p(\text{no collision}) = \left(1 - \frac{1}{365}\right) \cdot \left(1 - \frac{2}{365}\right)$$

- Similarly, when there are four people, the probability that no two people have the same birth date is:

$$p(\text{no collision}) = \left(1 - \frac{1}{365}\right) \cdot \left(1 - \frac{2}{365}\right) \cdot \left(1 - \frac{3}{365}\right)$$

- Finally, if there are p people, the probability that no two people have the same birth date is,

$$p(\text{no collision}) = \left(1 - \frac{1}{365}\right) \cdot \left(1 - \frac{2}{365}\right) \cdot \left(1 - \frac{3}{365}\right) \dots \left(1 - \frac{p-1}{365}\right)$$

$$p(\text{no collision}) = \prod_{i=1}^{p-1} \left(1 - \frac{i}{365}\right)$$

The Birthday paradox cntd..

- Finally, if there are p people, the probability that no two people have the same birth date is,

$$p(\text{no collision}) = \prod_{i=1}^{p-1} \left(1 - \frac{i}{365}\right)$$

Generalizing this equation for n people random values instead of 365 possible birth dates, the probability that no collision occurs if there are p random elements is ,

$$\prod_{i=1}^{p-1} \left(1 - \frac{i}{n}\right)$$

The equation further can be reduced by considering the Taylor Series:

$$e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \dots$$

If $x \ll 1$, then $e^{-x} \approx 1 - x$.

Therefore, the probability that no collision occurs if there are p random elements may be rewritten as:

$$p(\text{no collision}) \approx \prod_{i=1}^{p-1} e^{-\frac{i}{n}} \approx e^{-\frac{1}{n}} \cdot e^{-\frac{2}{n}} \cdot e^{-\frac{3}{n}} \cdot \dots \cdot e^{-\frac{(p-1)}{n}} \approx e^{-\frac{(1+2+3+\dots+p-1)}{n}} \approx e^{-\frac{p \cdot (p-1)}{2n}}$$

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- Finally, if there are p people, the probability that no two people have the same birth date is,

$$p(\text{no collision}) \approx e^{-\frac{p \cdot (p-1)}{2n}}$$

Now, the probability of that at least one collision occurs determined as,

$$p(\text{at least one collision}) \approx 1 - p(\text{no collision})$$

$$p(\text{at least one collision}) \approx 1 - e^{-\frac{p \cdot (p-1)}{2n}}$$

If ε (*epsilon*), is used to denote the probability that at least one collision occurs, then:

$$\varepsilon \approx 1 - e^{-\frac{p \cdot (p-1)}{2n}}$$

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$$\rightarrow \varepsilon \approx 1 - e^{-\frac{p \cdot (p-1)}{2n}}$$

$$\rightarrow e^{-\frac{p \cdot (p-1)}{2n}} \approx 1 - \varepsilon$$

$$\rightarrow -\frac{p \cdot (p-1)}{2n} \approx \ln(1 - \varepsilon)$$

$$\rightarrow p \cdot (p - 1) \approx -2n \cdot \ln(1 - \varepsilon)$$

$$\rightarrow p^2 - p \approx 2n \cdot \ln\left(\frac{1}{1-\varepsilon}\right)$$

For large values of p , i.e. a large number of random elements to be tested to find a collision, p^2 is significantly large than p .

Therefore,

$$p^2 \approx 2n \cdot \ln\left(\frac{1}{1-\varepsilon}\right)$$

Therefore, the probability that at least one collision occurs is ,

$$p \approx \sqrt{2n \cdot \ln\left(\frac{1}{1-\varepsilon}\right)}$$

Example1

- How many people must be present so that there is a 50% probability that at least two of the people have the same birth date if leap year ignored such that February 29th is not viable birth date, i.e. that $n=365$?

Solution:

Example2

The birthday paradox can be extended to Hash functions to determine how many random outputs must be generated such that the probability of a collision occurring is a specified percentage.

- How many random outputs are required so that there is a 50% probability of a collision occurring if an 80-bit Hash Function is used?

Solution: